Physics-Guided Probabilistic Glucose Forecasting in Type 1 Diabetes: An Integrity-Checked Multi-Horizon Evaluation with a Stable Patient-Specific Insulin-Sensitivity Estimate

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***Abstract*—Background and objective: Glucose dynamics in type 1 diabetes are nonlinear, delayed, and patient-specific, driven by meals and insulin. This study develops a physics-guided probabilistic forecaster that combines deep sequence learning with mechanistic physiological modeling, and tests what the physics adds beyond accuracy under an integrity-checked protocol.**

Methods: On OhioT1DM (12 subjects), only windows whose 30–120 min targets were real CGM readings were used (26,498 test windows). A Transformer encoder was coupled to a differentiable Bergman minimal model, with patient-specific parameter estimation and a non-crossing quantile head. Evaluation was personalized and leave-one-subject-out (LOSO). A pre-specified ablation tested the physics contribution, and pre-specified checks tested whether insulin sensitivity (S_I) was subject-specific, stable, and dose-consistent. Post hoc analyses repeated the physics comparison over five seeds and across the two OhioT1DM cohorts.

Results: Personalized forecasts had 19.4–21.2% lower RMSE than persistence at 30–120 min, and at 30 min all 12 subjects beat persistence in MAE under both protocols. The pre-specified single-seed test was not significant (Holm $p = 0.105$); over five seeds, physics lowered MAE by 0.17–0.99 mg/dL (Holm $p \leq 0.010$). Across cohorts, physics lowered MAE in all 12 test subjects by 1.7–7.4 mg/dL, though no test survived Holm correction. S_I differed between subjects (CV 35.8%), was stable within each (ICC 0.89), and was lower in subjects using more insulin ($\rho = -0.84$), a relation that did not hold under LOSO. A 10th-percentile alarm caught 91.8% of hypoglycemic readings (median forecast: 55.1%), yet a tuned persistence alarm detected as many episodes (38 of 56) at a similar false-alarm rate.

Conclusion: Unlike evaluations of physics-guided forecasters based on accuracy alone, this study shows what physics adds: a stable insulin-sensitivity estimate, dose-consistent under personalized evaluation, and small accuracy gains that grow under cross-cohort transfer, but no better hypoglycemia detection than a tuned persistence alarm.

***Index Terms*—continuous glucose monitoring, type 1 diabetes, physics-guided machine learning, Bergman minimal model, probabilistic forecasting, hypoglycemia, insulin sensitivity**

I. Introduction

A. Background

Type 1 diabetes (T1D) is a disease involving the immune-mediated destruction of pancreatic β-cells. Individuals with T1D therefore require lifelong exogenous insulin replacement and glucose monitoring, and disease-specific education, diet, and lifestyle modification remain cornerstones of management alongside insulin therapy. Glucose–insulin dynamics, however, are governed by nonlinear, patient-specific physiology, as described mechanistically through the Bergman minimal model [1]. Despite advancements in continuous glucose monitoring (CGM) hardware, it remains very difficult to build a system that converts raw diabetes data into personalized, actionable medical recommendations [2].

This has driven the development of machine-learning approaches for CGM forecasting, progressing from recurrent networks to attention-based Transformer architectures [3-6], which can achieve strong short-horizon accuracy but offer clinicians little interpretability [7-8]. This limitation motivated a shift toward physiology-aware approaches that add a physics-consistency penalty derived from mechanistic models, following the Physics-Informed Neural Network (PINN) paradigm [9-12]. However, some of these models were validated only in simulation [10]. More recent systems introduced patient-specific digital twins [13-15] and prospective evaluation [16-17]. Yet translating predictions into safe, actionable recommendations remains challenging, with reinforcement-learning approaches still considered insufficiently clinically safe [18].

B. Literature Review

Explainable short-horizon hypoglycemia/hyperglycemia prediction models added SHAP-based feature attributions to opaque risk classifiers, demonstrating interpretable patient-level prediction [7]. Other work used SHAP attributions to guide bounded, one-step behavioural recommendations on OhioT1DM, though the approach remained exploratory and was not considered clinically safe [18]. Explainable postprandial glucose modeling attributed meal-response forecasts to specific inputs, achieving personalized, transparent prediction, though remaining purely data-driven with no physiological constraint [19].

Physiology has also been embedded directly into the learning objective. Early PINN-based glucose predictors added a physics-consistency penalty to sequence learning, improving short-term in-silico prediction, though validation stayed simulator-only [10]. Hybrid physiology and deep-learning

architectures embedded glucose–insulin dynamics into both architecture and training, achieving reliable predictions under limited real-subject data, but remained forecasting models rather than virtual patients [11]. Physics–LSTM hybrids addressed recurrent black-box models' inability to distinguish meal and insulin effects, though still without a full virtual-patient representation [12].

Physiology-based digital twins have been built for in-silico therapy testing and counterfactual simulation [13], [14], used as synthetic data generators [20], validated against real exercise data [15], and tested in randomized trials of decision support and periodic re-optimization of automated insulin delivery [16], [17]; these systems optimize therapy rather than forecast glucose continuously, which is the task addressed here.

In parallel, attention-based Transformer forecasting on OhioT1DM outperformed classical machine-learning baselines (XGBoost, 1D-CNN) for short-horizon prediction but remained a plain sequence model with limited interpretability [3]. Other work added interpretable multi-horizon attention over temporal and static covariates [4]. A hybrid Transformer–LSTM architecture improved short- and medium-horizon accuracy, though horizon coverage stopped short of 120 minutes [5], while long-horizon hybrids reached 120 minutes with richer clinical inputs but remained purely data-driven [6]. Personalized federated learning allowed models to be trained across multiple sites without sharing patient data, but did not consider on-device efficiency or interpretability [21]. Lightweight Transformers made edge deployment more practical, although they lacked built-in uncertainty estimation and clear interpretability [22]. A Temporal Fusion Transformer improved RMSE by $\geq$13% over earlier baselines and provided feature and temporal importance with calibrated uncertainty, but its attention patterns were not validated against physiological ground truth [23]. Self-supervised Transformer pretraining learned reusable glucose representations for different downstream tasks [24]. Retrieval-augmented forecasting used similar historical episodes rather than raw sequences alone, but still lacked mechanistic grounding, SHAP-based explanations, and prospective clinical validation [25].

Three questions therefore remain open. First, how much of the accuracy reported on OhioT1DM survives an evaluation with verified horizons, a persistence baseline, and subjects rather than overlapping windows as the unit of analysis? Second, does embedding physiology improve point accuracy, or does its value lie in the parameters it produces, and are those parameters valid? Third, can a predictive distribution detect hypoglycemia that an accurate median forecast misses?

Relative to the physics-informed and hybrid forecasters above [10], [11], [12] and [33], this work does not propose a new physiological model. Its contribution is the combination, within one evaluation, of (i) real-CGM-only targets at verified 30–120-min horizons with subjects as the unit of analysis; (ii) a pre-specified ablation, a multi-seed test, and a cross-cohort transfer test that separate the contribution of physics from that of richer inputs; (iii) pre-specified stability and dose-consistency checks of a learned patient parameter (S_I); and (iv) quantile forecasts whose hypoglycemia alarms are compared with a simple persistence alarm, and point accuracy reported under both personalized and leave-one-subject-out protocols.

Table I situates this study among OhioT1DM forecasting approaches, grouped into non-physics learned or statistical sequence models and physics-informed hybrids; the proposed method belongs to the latter group.

C. Objectives and Contributions

- To develop a hybrid physics-guided forecaster that couples a Transformer with Bergman minimal-model constraints for 30–120-minute glucose forecasting, as the basis for testing what physiological structure adds.
- To evaluate the framework under an integrity-checked protocol, using only real CGM targets and subjects as the unit of analysis, and to test it under both personalized and subject-disjoint (leave-one-subject-out) protocols on identical windows.
- To separate the value of physics from accuracy gain through a pre-specified ablation, a five-seed replication, and a cross-cohort transfer test, rather than reporting accuracy alone.
- To estimate patient-specific insulin sensitivity from routine CGM, insulin, and meal data, and to verify through pre-specified checks that it is individual, stable, and consistent with total daily insulin dose.
- To quantify forecast uncertainty and evaluate lower-quantile hypoglycemia alarms against a validation-tuned persistence alarm.

The paper is structured as follows. Section I introduces the background, related literature, and objectives. Section II details the methodology, including data acquisition and model development. Section III presents the experimental results and model evaluation. Section IV discusses the findings, Section V covers limitations and future work, and Section VI concludes. The appendix lists the patient-parameter vector; full notation tables are in the Supplementary Material.

II. Methodology

A. Integrity Requirements

Glucose-forecasting pipelines on OhioT1DM are exposed to several integrity pitfalls that can inflate or invalidate reported accuracy: windows that span recording gaps, so that the nominal horizon differs from the true one; forward-filled or interpolated CGM values used as prediction targets; random splits of strongly overlapping windows, which place near-duplicate samples in both training and test sets; overlapping windows, rather than subjects, used as the unit of statistical analysis; and no persistence baseline, without which absolute error values cannot be interpreted. Published no-change baselines for OhioT1DM [38-39] make the last pitfall straightforward to check.

The pipeline used here was built to exclude each pitfall by design: each became a design constraint or one of 234 automated tests. Evaluation begins with persistence, subjects rather than windows are the unit of analysis, and every table is regenerated from stored predictions.

TABLE I
LANDSCAPE COMPARISON OF OHIOT1DM BLOOD-GLUCOSE FORECASTING STUDIES BY ARCHITECTURE FAMILY (30/60/90/120-MINUTE HORIZONS)

Reference	Methodology	Cohort	RMSE@30	MAE@30	RMSE@60	MAE@60	RMSE@90	MAE@90	RMSE@120	MAE@120
Non-physics learned/statistical sequence models (6 subjects, official split)										
[74]	CNN/LSTM	2020	17.45	11.22	33.67	23.25	not reported†		not reported†	
[37]	Neural physiological encoder + LSTM	2018	17.80	–	–	–	not reported†		not reported†	
[73]	N-BEATS + BiLSTM	2020	18.22	12.83	31.66	23.60	not reported†		not reported†	
[72]	LSTM	2020	18.23	14.37	31.10	25.75	not reported†		not reported†	
[75]	GAN	2020	18.34	13.37	32.21	24.20	not reported†		not reported†	
[71]	NN-EIM	2020	18.63	10.08	32.27	17.69	not reported†		not reported†	
[26]	Stacked regression + activity	2020	18.99	13.73	33.39	25.04	not reported†		not reported†	
[66]	MS-LSTM	2020	19.05	13.50	32.03	23.83	not reported†		not reported†	
[27]	Multi-lag stacking	2020	19.21	13.93	33.65	25.31	not reported†		not reported†	
[34]	Dilated CNN (GluNet)	2018	19.28	–	31.83	–	not reported†		not reported†	
[28]	Latent-variable statistical	2020	19.37	13.76	32.59	24.64	not reported†		not reported†	
[29]	Multi-class LSTM (+2018 data)	2020	19.40	13.90	33.40	25.00	not reported†		not reported†	
[35]	SVR (multivariate)	2018	19.59	13.49	31.54	22.86	not reported†		not reported†	
[30]	Multitask CRNN	2020	19.79	13.62	33.73	24.54	not reported†		not reported†	
[36]	Gaussian process	2018	20.01	–	31.97	–	not reported†		46.24	–
[31]	Online ARMA + residual net	2020	20.03	14.52	34.89	24.61	not reported†		not reported†	
[32]	Personalised interpretable LSTM	2020	20.20	14.74	34.19	25.98	not reported†		not reported†	
Physics-informed + other-architecture hybrid (6 subjects, official split)†										
[33]	Physiological model + ANN	2018	19.33	–	31.72	–	not reported†		not reported†	
–	**Current Work: Hybrid physics-guided Transformer (Bergman penalty + prior)**	2020	**18.74**	**13.24**	**31.16**	**22.55**	**39.52**	**29.14**	**45.39**	**34.03**
–	**Current Work (same model)**	2018	**18.93**	**12.92**	**29.89**	**21.42**	**37.69**	**27.70**	**43.53**	**32.44**
–	*Persistence baseline (ours)*	2020	24.22	17.45	40.11	29.51	51.15	38.34	58.98	44.83
–	*Persistence baseline (ours)*	2018	22.52	16.28	36.19	26.92	46.22	34.76	53.89	40.97

† Cohort: OhioT1DM test subjects on the official split (2018: 559, 563, 570, 575, 588, 591; 2020: 540, 544, 552, 567, 584, 596); results are comparable only within a cohort. Some studies score forecasts against interpolated or extrapolated test values; the current work uses only recorded CGM readings. "Not reported"/–: not published for that horizon or metric (the BGLP challenges required only 30- and 60-min forecasts). All-subject results: Table IV. Most comparators are BGLP challenge workshop papers.

B. Dataset and Temporal Integrity

OhioT1DM contains 12 subjects in the 2018 and 2020 cohorts and 166,443 CGM observations sampled nominally every five minutes, together with basal and bolus insulin, carbohydrate, activity, sleep, and wearable channels [40-41]. The official split follows the same subjects forward in time and therefore measures personalized forecasting, not cross-subject generalization. The 2020 cohort's first test hour is excluded according to its challenge protocol; the 2018 cohort is not modified in this way, so results are compared only within a cohort.

All streams are aligned to an exact five-minute grid without dropping rows. Missing CGM runs of at most ten minutes may be interpolated for input construction, but interpolated values are never accepted as targets. A sequence is emitted only if its 24-step (two-hour) input satisfies the coverage rule and every target is a real observation exactly $h \in \{30, 60, 90, 120\}$ minutes after the anchor. The scaler is fit only on training-fold slots, with each physical slot counted once rather than once per overlapping window.

From 187,780 candidate windows, 141,100 (75.1%) survive: 36,395 are rejected for incomplete inputs and 10,285 for unavailable real-valued targets. Both evaluation protocols use the same 26,498 test windows. Consecutive windows share 23 of their 24 input steps, so the raw window count overstates the information available. Each window, together with its 120-min target, spans 48 five-minute steps (24 input plus 24 forecast), so windows share observed data unless they are at least 48 steps apart. Dividing the approximately 135,000 retained five-minute slots across the whole cohort (the 141,100 retained windows overlap and share these slots) by this block length gives

$$n_{\text{eff}} \approx \frac{135{,}000 \text{ slots}}{48 \text{ steps}} \approx 2{,}800, \tag{1}$$

a coarse approximation. An autocorrelation-based estimate [42], using the mean integrated autocorrelation time of 53.6 five-minute steps across subjects, gives $n_{\text{eff}} \approx 2{,}500$–$3{,}100$, consistent with Eq. (1). Both figures describe the whole cohort. The test set alone, on which all reported metrics and the hypoglycemia analysis rest, corresponds to about $26{,}498/48 \approx 552$ effectively independent windows, or about 494 using the 53.6-step autocorrelation time.

C. System Overview

Fig. 1 outlines the full pipeline from raw recordings to reported outcomes. Data collected from all twelve OhioT1DM subjects, comprising continuous glucose readings, insulin delivery, and meal logs, are first aligned to a uniform five-minute time grid; genuine gaps in recording are preserved, and only short gaps of at most ten minutes may be interpolated for model inputs, never for forecast targets. From this aligned record, two parallel processes follow. The grid-aligned insulin and meal signals are passed through a simplified physiological model of glucose regulation, based on the Bergman minimal model together with subcutaneous insulin and gut absorption compartments, which yields time-varying estimates of insulin on board, carbohydrate on board, remote insulin action, and

the rate of glucose appearance. These physiological estimates are then combined with the glucose signal itself, insulin and meal timing, and time-of-day information to form a single 35-feature representation at each time step.

This feature representation is grouped into two-hour input windows, and windows are retained only when both the input span and the corresponding forecast target are fully intact, so that no window silently spans a genuine data gap. The resulting windows are evaluated under two complementary protocols: an official, personalized setting in which each subject's earlier data are available for training, and a twelve-fold leave-one-subject-out setting in which the model is evaluated on subjects it has never encountered. Each window is then processed by a compact Transformer encoder, which produces two outputs in parallel: an estimate of the subject's individual physiological parameters, and a probabilistic glucose forecast expressed as a median trajectory bounded by non-crossing lower and upper quantiles.

These two outputs are not independent. The estimated physiological parameters are used to re-simulate the same mechanistic equations introduced earlier, and the resulting trajectory is compared against the network's own forecast at multiple points across the two-hour horizon; the discrepancy between the two constitutes a physics-consistency signal that is folded back into training, alongside a regularization term that keeps the estimated parameters within physiologically plausible bounds. In this way, the mechanistic model contributes to the pipeline twice: once as an input feature, and once as a differentiable constraint on the learned forecast. Because physics enters training both as a residual penalty and as a gated prior, we refer to the model as a *hybrid physics-guided* forecaster (penalty + prior); "PINN" is used only for the literature paradigm. Finally, forecasts from both evaluation protocols are scored on a per-subject basis and summarized across the cohort, using both conventional accuracy metrics and clinically oriented measures of hypoglycemia detection.

D. Mechanistic Patient Model

The physiological core extends the Bergman minimal model and its insulin-sensitivity index [1], [43], as given in Eqs. (2)–(4):

$$\dot{G} = -(p_1 + X)\, G + p_1 G_b + \frac{R_a(t)}{V_G}, \tag{2}$$

$$\dot{X} = -p_2 X + p_3 (I - I_b), \tag{3}$$

$$\dot{I} = -nI + \frac{1000}{V_I} \cdot \frac{S_2}{t_{\max,I}}. \tag{4}$$

Here G is glucose, I plasma insulin, X remote insulin action, and subscript b denotes basal state. All rates are per minute. Basal subtraction in (3) is essential: at $I = I_b$, $X^* = 0$ and $G = G_b$ is an equilibrium. Driving the equation with total insulin instead would create a spurious downward glucose attractor.

The patient-specific insulin sensitivity is the steady-state gain, as given in Eq. (5):

$$S_I = \frac{p_3}{p_2}, \qquad X_{ss} = S_I (I - I_b), \tag{5}$$

with units $\mathrm{mL}\,\mu\mathrm{U}^{-1}\mathrm{min}^{-1}$. Parameters are mapped by sigmoid functions into physiologically sourced intervals informed by minimal-model studies in insulin-dependent diabetes [44] and regularized toward population anchors. Consequently, plausible range alone is not evidence of identifiability; external and stability checks are required.

Subcutaneous insulin absorption follows a two-compartment model [45], as given in Eqs. (6)–(8):

$$\dot{S}_1 = u_{\text{ins}}(t) - \frac{S_1}{t_{\max,I}}, \tag{6}$$

$$\dot{S}_2 = \frac{S_1}{t_{\max,I}} - \frac{S_2}{t_{\max,I}}. \tag{7}$$

For an impulse dose D, insulin on board is

$$S_1 + S_2 = D\, e^{-t/t_{\max,I}} \left(1 + \frac{t}{t_{\max,I}}\right), \tag{8}$$

which starts at D and decreases monotonically. Carbohydrate absorption follows compartmental meal models used in T1D simulation [46-48] and is represented by stomach and gut states, as given in Eqs. (9)–(11):

$$\dot{Q}_{\text{sto}} = -k_{\text{gri}} Q_{\text{sto}} + u_{\text{carb}}(t), \tag{9}$$

$$\dot{Q}_{\text{gut}} = k_{\text{gri}} Q_{\text{sto}} - k_{\text{abs}} Q_{\text{gut}}, \tag{10}$$

$$R_a(t) = f\, k_{\text{abs}} Q_{\text{gut}}(t), \tag{11}$$

so carbohydrate on board is $Q_{\text{sto}} + Q_{\text{gut}}$ and absorbed mass is conserved. For implementation, the non-glucose compartments are collected, as given in Eq. (12):

$$\mathbf{z} = [S_1, S_2, I, Q_{\text{sto}}, Q_{\text{gut}}, X]^{\mathsf{T}}, \quad \mathbf{u} = [u_{\text{ins}}, u_{\text{carb}}, I_b]^{\mathsf{T}}. \tag{12}$$

With parameters held constant over one grid interval, $\dot{\mathbf{z}} = A\mathbf{z} + B\mathbf{u}$ has the exact zero-order-hold update (a standard control-theory discretization technique, not attributed to a specific glucose-modeling source), as given in Eq. (13):

$$\mathbf{z}_{k+1} = A_d \mathbf{z}_k + B_d \mathbf{u}_k, \qquad \begin{bmatrix} A_d & B_d \\ 0 & I \end{bmatrix} = \exp\left(\Delta \begin{bmatrix} A & B \\ 0 & 0 \end{bmatrix}\right). \tag{13}$$

This matrix-exponential formulation preserves the affine basal term through the explicit I_b input and avoids the accumulated integration error of a coarse explicit solver.

E. Feature Contract and Forecasting Network

Each timestep contains 35 named features in six groups: glucose level, multiscale rates of change and rolling statistics; delivered insulin and timing; meals and timing; mechanistic states (IOB, COB, plasma insulin, remote action, and glucose appearance); cyclic time; and context/sensor variables. A two-hour window $X \in \mathbb{R}^{24\times 35}$ is standardized using training-only statistics.

The forecasting network is deliberately small (approximately 79,000 parameters) because the effective sample size is limited. A linear projection and positional encoding feed a Transformer encoder [49]. Attention pooling, Eq. (14), gives

$$z = \sum_{t=1}^{24} \alpha_t h_t, \qquad \alpha_t = \frac{\exp(w^\top h_t)}{\sum_j \exp(w^\top h_j)}, \tag{14}$$

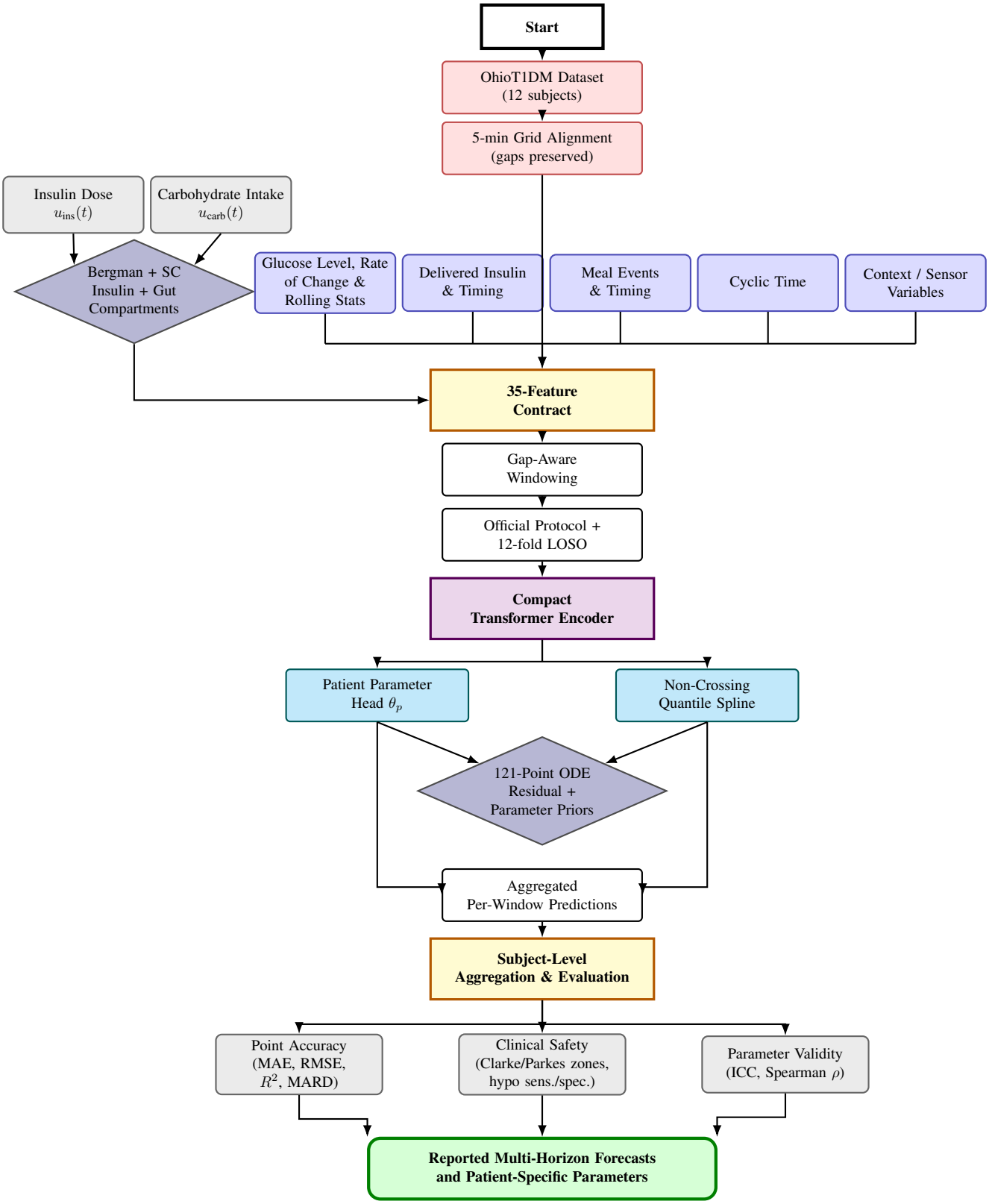


Fig. 1. *Integrity-checked forecasting pipeline of the hybrid physics-guided model. OhioT1DM glucose and related patient data are aligned to five-minute intervals and processed through two parallel paths: a deep-learning model and a physiological model. The deep-learning model predicts future glucose levels and estimates patient-specific parameters, while the physiological model represents insulin and glucose dynamics. Physiological information is used both as an input to the model and as a training constraint to improve prediction consistency.*

avoiding the equal weighting of old and recent measurements induced by mean pooling.

One head estimates the ten-component subject parameter vector θ_p. A second predicts a cubic B-spline trajectory, as given in Eq. (15):

$$G_\theta(t) = G_0 + \sum_{k=1}^{K} c_k\big(B_k(t) - B_k(0)\big), \tag{15}$$

which anchors every trajectory at the last observed glucose G_0. Median knot values are unconstrained, whereas lower and upper quantiles are generated by positive softplus offsets in glucose units; quantiles therefore cannot cross. A learned trust gate $g = \sigma(\gamma)$ controls the contribution of the mechanistic prior, with $g \approx 0.10$ at initialization. Predictions at all four horizons are evaluated from the spline.

F. Training Objective

The data objective combines Huber's robust loss [50] for the median with the pinball loss used in quantile regression [51] for lower and upper quantiles, as given in Eq. (16):

$$\mathcal{L}_{\text{data}} = \mathcal{L}_{\text{Huber}}(y, \hat{y}_{0.5}) + \sum_{q\in\{0.1,0.9\}} \rho_q(y - \hat{y}_q), \tag{16}$$

where $\rho_q(u) = u(q - \mathbb{1}_{u<0})$. Physics residuals are evaluated at 121 collocation points over the two-hour forecast and normalized by physiological scales, following the residual-minimization principle of PINNs [9]. The full objective is given in Eq. (17):

$$\mathcal{L} = \mathcal{L}_{\text{data}} + \lambda_{\text{phys}}\mathcal{L}_{\text{phys}} + \lambda_{\text{prior}}\mathcal{L}_{\text{prior}} + \lambda_{\text{tc}}\mathcal{L}_{\text{temporal}}. \tag{17}$$

Closed-form state updates replace a memory-intensive differentiable ODE solver. Adaptive loss weighting was motivated by work on gradient imbalance and uncertainty-based multi-objective learning [52-53]. The A7 configuration, selected on validation data rather than the test set, applies physics from epoch one; a data-first curriculum was rejected because validation optima repeatedly occurred before the curriculum became selectable.

G. Evaluation and Statistical Analysis

The official temporal protocol uses each subject's earlier data for training. LOSO trains on 11 subjects and evaluates the held-out subject on the identical test windows. Metrics are computed per subject and summarized as mean $\pm$ SD. Persistence predicts $\hat{G}(t+h) = G(t)$. Skill is $1 - \text{RMSE}_{\text{model}}/\text{RMSE}_{\text{persistence}}$, computed from the cohort-mean RMSEs.

Clinical evaluation includes Clarke and Parkes zones [54-56], consensus CGM ranges [57], coefficient-of-variation ratio, and Kovatchev low/high blood-glucose risk indices [58-59], together with hypoglycemia sensitivity, specificity, precision, and quantile calibration. Paired model comparisons use the Wilcoxon signed-rank test [60] across 12 subjects with Holm multiplicity correction [61]; bootstrap intervals resample subjects, never overlapping windows. Test–retest reliability of S_I is measured with one-way ICC(1) [62], and concurrent validity uses Spearman rank correlation [63].

H. Pre-specified Outcomes and Falsification Criteria

The outcomes and criteria below were fixed on 25 July 2026, before any test-set evaluation. The primary outcome is 30-minute MAE under the official protocol for the physiology-focused branch. Secondary outcomes include all horizons under both protocols, persistence skill, Clarke, Parkes, and PRED-EGA distributions, the ablation matrix, quantile calibration, and the three S_I validation checks. Table II records the criteria that were fixed in advance; unmet criteria are reported as failures rather than redefined after inspection.

No asymmetric hypoglycemia/hyperglycemia point-loss term is permitted, because such a term would directly optimize the safety table later used as evidence. The test set is evaluated only after model selection, and the official protocol is never described as subject-disjoint generalization.

After the primary results were known, nine secondary analyses of the physics contribution were written down and time-stamped before they were run (the record is in the code repository): the horizon trend of the A0–A7 gap; strata by post-meal period, target glycemic range, and glucose rate of change; physiological plausibility of the forecast trajectory; reduced training data (25% and 50% of subjects, three seeds); seed variance; transfer between the 2018 and 2020 OhioT1DM cohorts; and five-seed hypoglycemia alarm performance. They are reported in full, regardless of direction, with Holm correction within each family; they are not part of the 25 July 2026 protocol.

I. Reproducibility, Hardware, and Latency

Models were implemented in PyTorch 2.10 (CUDA 12.8) and trained with AdamW (learning rate 5×10^{-4}, weight decay 0.03, batch size 256, gradient clipping at L_2 norm 1.0) for at most 70 epochs, with early stopping on 30-min validation MAE (patience 10 for the ablation arms and all multi-seed and secondary-analysis runs, and 8 for the final quantile-head model, whose best epoch was 3 of 11). Validation used the final 15% of each subject's time-ordered training windows, separated from training by a purge gap of one input window plus the longest horizon; under LOSO, each training subject contributed its own purged validation tail, and the held-out subject was used only for testing. The hyperparameter search covered encoder depth $\in \{2, 4\}$, $d_{\text{model}} \in \{64, 128\}$, attention heads $\in \{4, 8\}$, $d_{\text{ff}} \in \{128, 512\}$, dropout $\in \{0.1, 0.2\}$, pooling (attention, mean, or last), and physics-loss weighting (fixed $\lambda_{\text{phys}} \in [0.01, 1.0]$ vs. Kendall homoscedastic uncertainty weighting); validation selected 2 layers, $d_{\text{model}} = 64$, 4 heads, $d_{\text{ff}} = 128$, dropout 0.2, attention pooling, and Kendall weighting.

Experiments ran on a single Linux machine with an NVIDIA GeForce RTX 2050 Laptop GPU (4 GB) and an Intel Core CPU. Across the seven primary training runs (ablation arms A0–A4 and A7, and the final quantile-head model), training stopped after 11–20 epochs (mean 16.3), at 10.7–23.8 s per epoch, or 2.7–6.4 min per run. Inference latency for one window (batch size 1, 100 forward passes) was 15.25 ± 4.07 ms on GPU (median 13.21 ms, 95th percentile 25.07 ms) and 6.33 ± 0.18 ms on CPU (median 6.31 ms),

well within the 5-minute CGM sampling interval. Code, data-processing pipelines, and configurations are available at https://github.com/sammyyakk/digital-twin.

J. Feature Attribution Method

Integrated gradients (IG) [64] uses the standardized training mean as its zero baseline, as given in Eq. (18):

$$\mathrm{IG}_i(x) = (x_i - x_i') \int_0^1 \frac{\partial f(x' + \alpha(x - x'))}{\partial x_i}\, d\alpha. \qquad (18)$$

Because IG shows an unresolved completeness violation, every interpretation is cross-checked with model-agnostic permutation importance, following the perturbation principle popularized for random forests [65].

III. Results

A. Baseline Validation and Data Accounting

Persistence RMSE on the 2018 cohort is 22.52 ± 2.56 mg/dL at 30 minutes and 36.19 ± 3.26 mg/dL at 60 minutes, within 0.02 mg/dL of both independently published no-change baselines at 30 minutes and within 0.41 mg/dL of the published 60-minute value[38-39]. This close agreement validates the complete evaluation pipeline, including data parsing, five-minute grid alignment, horizon construction, persistence-target generation, RMSE computation, and subject-level aggregation (Table III). The retained-window accounting further confirms that the reported metrics are computed on the intended 26,498 common test windows.

B. Official-Protocol Forecast Accuracy

Table IV and Fig. 2 show that the model improves on persistence at every horizon. All 12 subjects improve at 30 minutes; subject-level MAE skill ranges from 12.2% to 31.6%. The decline in R^2 with horizon is expected because future meals, boluses, and activity are absent from the input.

C. Comprehensive Baseline Comparison on Identical Windows

Table V compares eleven models on the same 26,498 test windows from all 12 subjects under the official protocol:

1) **Persistence:** zero-order hold, $\hat{G}_{t+h} = G_t$.
2) **Bergman ODE, population parameters:** forward integration from the basal steady state with literature parameters [1], [43], without neural correction.
3) **Bergman ODE, per-subject fit:** parameters fitted per subject by nonlinear least squares/MAP on historical calibration intervals.
4) **AR(6):** linear autoregression on the last six CGM values (30-min lookback).
5) **ARX(6):** AR(6) with added insulin and carbohydrate inputs.
6) **ARX(6) + IOB/COB:** ARX(6) with mechanistic insulin- and carbohydrate-on-board added.
7) **Ridge (35 features):** ridge regression on the full 35-feature input used by A0 and A7.
8) **LSTM:** recurrent model on the 26 non-mechanistic features.
9) **Transformer, $A0_{\text{no-mech}}$:** the hybrid model's backbone trained on the same 26 features.
10) **Transformer, A0:** the same backbone with all 35 features, including mechanistic insulin- and carbohydrate-on-board.
11) **Hybrid physics-guided, A7:** A0 plus the Bergman prior, learned residual correction, and parameter head.

Mechanistic models alone performed worst: with population parameters the ODE drifted to a 30-min MAE of 29.70 mg/dL (54.18 mg/dL at 120 min), and per-subject fitting improved this only to 25.80 mg/dL (48.43 mg/dL at 120 min). AR(6) outperformed both non-mechanistic neural baselines at every horizon (30-min MAE 14.39 vs. 15.41 mg/dL for the LSTM and 14.96 mg/dL for $A0_{\text{no-mech}}$), and adding insulin and meal inputs (ARX) lowered MAE to 13.97 mg/dL. Adding mechanistic features to the Transformer reduced 30-min MAE by 1.45 mg/dL (14.96 to 13.51 mg/dL). To separate the contribution of richer inputs from that of the architecture, ridge regression on all 35 features and ARX(6) with IOB/COB were fitted on the same windows; they reached 30-min MAEs of 13.56 and 13.51 mg/dL, matching A0. At 30 min, therefore, most of the gain over the non-mechanistic neural baselines comes from the richer, mechanistically derived inputs, and A7 was 0.26–0.31 mg/dL lower than these linear models. The gap widened with horizon: A7's MAE was 1.34, 2.34, and 3.09 mg/dL below ridge regression at 60, 90, and 120 min. Differences from these two linear baselines were not tested for significance. A7 had the lowest MAE at every horizon in this comparison (13.25 mg/dL at 30 min).

D. Cross-Subject Generalization

At 30 minutes, LOSO remains better than persistence in all 12 subjects by MAE (10 of 12 by RMSE), but is consistently worse than the personalized protocol. As Fig. 3 shows, the personalization gap grows with horizon, consistent with increasing dependence on individual insulin kinetics and routine.

E. Cross-Cohort Transfer

OhioT1DM contains two cohorts recorded with different sensors and pumps (2018: subjects 559, 563, 570, 575, 588, 591; 2020: 540, 544, 552, 567, 584, 596). As a secondary analysis, A0 and A7 were each trained with five seeds on one cohort only, with validation data from the same cohort, and tested on the official test files of the other cohort. This is cross-cohort transfer within one dataset, not external validation. Both models were markedly less accurate than under the personalized protocol (30-min MAE 16.1–18.7 vs. 13.4–13.5 mg/dL), but A7 degraded less: it had lower MAE than A0 in all six test subjects at every horizon in both directions, by 1.7–4.8 mg/dL for 2018→2020 and 2.6–7.4 mg/dL for 2020→2018 (Table VI, Fig. 4); RMSE showed the same pattern. With six subjects per direction the smallest attainable Wilcoxon p is 0.031, and no test survived Holm correction across the 16 comparisons ($p_{\text{Holm}} = 0.5$), so the result is consistent in sign but statistically weak. It suggests that the physiological

TABLE II
PRE-SPECIFIED SUCCESS AND FALSIFICATION CRITERIA

Claim	Criterion fixed before test evaluation	Interpretation	Outcome	Ref.
Beats persistence	30-minute MAE below persistence in at least 10 of 12 subjects	Requires broadly distributed improvement, not a pooled gain driven by a few subjects.	Met: 12 of 12 subjects	[40-41], [39]
Field-competitive point accuracy	30-minute MAE $\leq$ 13.5 mg/dL	A field-relative target, explicitly not a clinical acceptability threshold.	Met: 13.08 (final model), 13.37 (5-seed mean)	[41], [54-55], [66]
Physics improves accuracy	Physics arm beats A0 at 30 minutes and survives paired Wilcoxon testing with Holm correction	If unmet, physics is not credited with point-accuracy improvement.	Not met: Holm $p = 0.105$ (post hoc, 5 seeds: -0.17 mg/dL, Holm $p = 0.010$)	[60-61]
S_I is subject-specific	Between-subject CV $> 10\%$	Rejects a parameter head collapsed to one population value.	Met: CV 35.8%	[1], [43]
S_I is stable	ICC(1) > 0.5 on disjoint within-subject blocks	Rejects a window-level nuisance variable that merely absorbs prediction error.	Met: ICC 0.890	[62], [67]
S_I is dose-consistent	Spearman $\rho < 0$ versus total daily dose, with a subject-bootstrap 95% CI excluding zero	Evaluated separately under personalized and LOSO protocols.	Met (official, $\rho = -0.839$); not met (LOSO)	[63], [68-69]

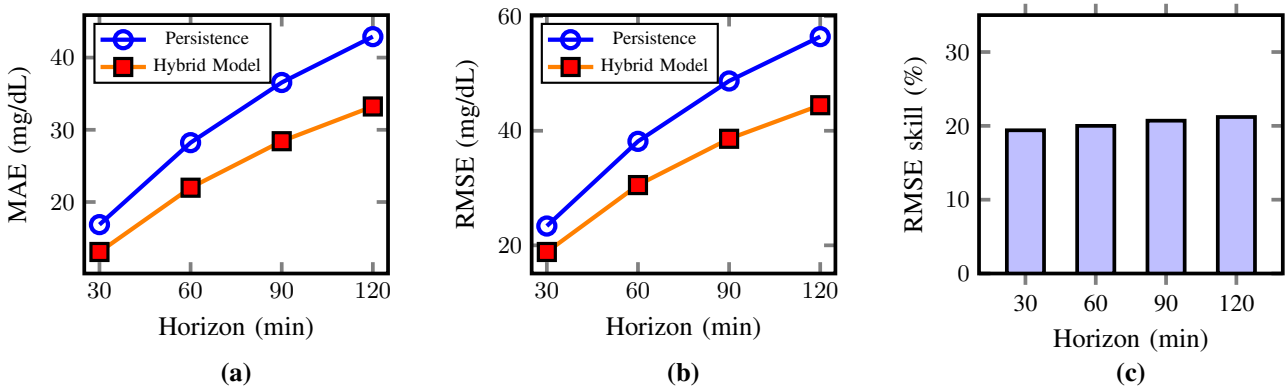


Fig. 2. *Official-protocol accuracy summary at 30/60/90/120 minutes. (a) and (b): MAE and RMSE for persistence versus the hybrid model both rise with horizon, and the gap between them widens. (c): the resulting RMSE skill stays essentially flat at 19–21% across all four horizons.*

TABLE III
WINDOW ACCOUNTING AFTER INTEGRITY FILTERING. ALL 12 SUBJECTS (2018 AND 2020 COHORTS); COUNTS ARE TOTALS ACROSS SUBJECTS, AND THE 26,498 TEST WINDOWS ARE SHARED BY BOTH EVALUATION PROTOCOLS.

Quantity	Value
Subjects / CGM observations	12 / 166,443
Candidate windows	187,780
Retained windows	141,100 (75.1%)
Incomplete input span	36,395
Missing real target	10,285
Common test windows	26,498
Effective independent windows, whole cohort	$\approx$ 2,800
Effective independent test windows	$\approx$ 552 ($\approx$ 494)

prior acts as a regularizer under distribution shift, although this design cannot separate that from improved physiological fidelity.

F. Physics Ablations

In the pre-specified single-seed ablation, no physics configuration significantly outperformed the no-physics A0 baseline after Holm correction (Table VII; Fig. 5). A7 was nominally best and improved 9 of 12 subjects, but its corrected

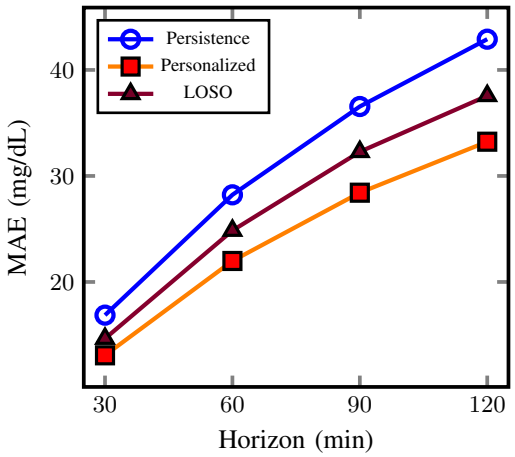


Fig. 3. *Matched-protocol comparison across 30, 60, 90, and 120-minute horizons. Persistence, personalized, and LOSO models are evaluated on identical windows. LOSO keeps a roughly constant MAE advantage of about 12% over persistence at every horizon, while its gap to the personalized model widens from 1.6 to 4.3 mg/dL.*

$p = 0.105$, so the pre-specified criterion was not met. Holm correction was applied across the five arm-versus-

TABLE IV
OFFICIAL PERSONALIZED PROTOCOL, ALL 12 SUBJECTS, FINAL MODEL (MEDIAN OF THE QUANTILE-HEAD RUN): MEAN ± SD ACROSS SUBJECTS (MG/DL UNLESS NOTED). TRAINED SEPARATELY FROM ABLATION ARM A7 (TABLES V–VII).

Horizon	Persistence MAE	Model MAE	Persistence RMSE	Model RMSE	RMSE skill	R^2	MARD
30 min	16.87 ± 1.92	**13.08 ± 1.73**	23.37	**18.84 ± 2.58**	19.4%	0.885	8.79%
60 min	28.22 ± 3.24	**21.98 ± 2.96**	38.15	**30.52 ± 3.98**	20.0%	0.700	14.76%
90 min	36.55 ± 4.00	**28.42 ± 3.89**	48.69	**38.60 ± 5.05**	20.7%	0.523	19.14%
120 min	42.90 ± 4.46	**33.23 ± 4.45**	56.43	**44.46 ± 5.62**	21.2%	0.368	22.37%

TABLE V
COMPREHENSIVE BASELINE COMPARISON ACROSS 12 SUBJECTS EVALUATED ON IDENTICAL TEST WINDOWS ($n = 26{,}498$). METRICS REPORTED AS MEAN ± SD ACROSS SUBJECTS (MG/DL). THE A7 ROW IS THE SINGLE-SEED ABLATION RUN (AS IN TABLE VII), NOT THE FINAL MODEL IN TABLE IV.

	30 min		60 min		90 min		120 min	
Model	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE
Persistence	16.87 ± 1.92	23.37 ± 2.85	28.22 ± 3.24	38.15 ± 4.58	36.55 ± 4.00	48.69 ± 5.58	42.90 ± 4.46	56.43 ± 6.21
Bergman ODE (population)	29.70 ± 9.67	39.67 ± 12.67	43.71 ± 12.84	57.08 ± 16.10	50.72 ± 13.66	65.26 ± 16.79	54.18 ± 13.78	69.20 ± 16.77
Bergman ODE (per-subject NLS)	25.80 ± 8.36	34.72 ± 10.45	38.09 ± 9.56	50.41 ± 11.78	44.70 ± 8.99	58.47 ± 11.01	48.43 ± 8.35	63.03 ± 10.29
AR(6)	14.39 ± 1.52	20.11 ± 2.65	25.52 ± 2.90	33.88 ± 3.90	33.18 ± 3.66	42.92 ± 4.62	38.30 ± 3.90	48.64 ± 4.92
ARX(6)	13.97 ± 1.51	19.61 ± 2.76	24.48 ± 2.86	32.67 ± 3.98	32.24 ± 3.47	41.82 ± 4.47	37.68 ± 3.69	48.00 ± 4.77
ARX(6) + IOB/COB	13.51 ± 1.50	19.09 ± 2.69	23.71 ± 2.86	31.86 ± 3.77	31.39 ± 3.51	41.04 ± 4.25	36.84 ± 3.74	47.22 ± 4.50
Ridge (35 features)	13.56 ± 1.55	19.10 ± 2.60	23.63 ± 3.07	31.80 ± 3.81	31.11 ± 3.89	40.94 ± 4.65	36.47 ± 4.18	47.04 ± 5.17
LSTM (non-mechanistic)	15.41 ± 2.16	21.46 ± 2.83	26.08 ± 3.87	35.24 ± 4.56	33.59 ± 4.77	44.97 ± 5.76	38.81 ± 5.30	51.54 ± 6.63
Transformer ($A0_{\text{no-mech}}$)	14.96 ± 1.82	21.08 ± 2.55	26.06 ± 3.15	35.56 ± 4.15	34.46 ± 3.81	46.08 ± 5.11	40.06 ± 4.09	53.03 ± 5.63
Transformer (all features, A0)	13.51 ± 1.63	19.38 ± 2.41	22.82 ± 2.88	31.61 ± 3.71	29.74 ± 3.74	40.32 ± 4.71	34.93 ± 4.20	46.70 ± 5.24
Hybrid physics-guided (A7)	**13.25 ± 1.69**	**19.06 ± 2.51**	**22.29 ± 2.93**	**30.89 ± 3.89**	**28.77 ± 3.75**	**39.04 ± 4.83**	**33.38 ± 4.21**	**44.78 ± 5.46**

TABLE VI
CROSS-COHORT TRANSFER: MAE (MG/DL), MEAN OVER FIVE SEEDS AND SIX TEST SUBJECTS. GAP = A0 − A7 WITH SUBJECT-BOOTSTRAP 95% CI; A7 WAS BETTER IN 6 OF 6 SUBJECTS IN EVERY ROW (WILCOXON $p = 0.031$, HOLM $p = 0.5$).

Direction	Horizon	A0	A7	Gap [95% CI]
2018→2020	30 min	18.38	16.64	1.74 [1.21, 2.30]
	60 min	31.48	27.80	3.68 [2.37, 5.08]
	90 min	40.57	35.77	4.80 [2.78, 6.94]
	120 min	46.39	41.57	4.82 [2.26, 7.47]
2020→2018	30 min	18.71	16.14	2.56 [2.24, 2.85]
	60 min	31.83	26.65	5.17 [4.59, 5.68]
	90 min	41.30	34.45	6.85 [5.95, 7.58]
	120 min	47.78	40.39	7.40 [6.31, 8.24]

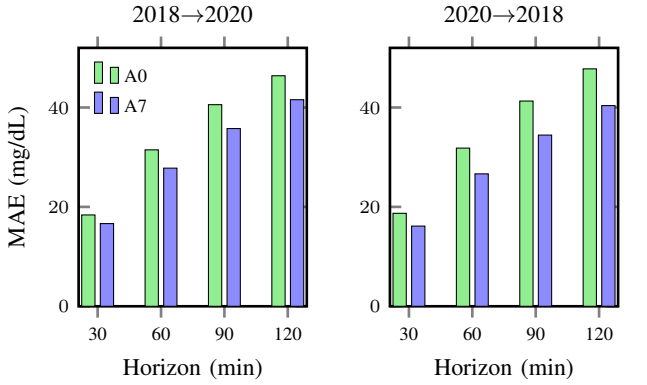


Fig. 4. *Cross-cohort transfer. MAE of A0 (no physics) and A7 (physics-guided) trained on one OhioT1DM cohort and tested on the other (five seeds, six test subjects per panel). A7 is lower in every subject and horizon, but no comparison survives Holm correction.*

A0 comparisons (A1–A4 and A7) on 30-min MAE. A post hoc 5-seed replication of A0 and A7, not part of the pre-specified plan (Section III-F1) found that, on seed-averaged per-subject errors, A7 has lower MAE at every horizon, by 0.17 mg/dL at 30 min and 0.99 mg/dL at 120 min (Holm $p \leq 0.010$ at each horizon). Per-patient parameters remain accuracy-neutral: A3 and population-fixed A4 differ by 0.07–0.23 mg/dL at 30/60 min, widening to 0.76 mg/dL at 120 min. In contrast with an earlier physiology-based OhioT1DM study that primarily reported forecast accuracy [33], this ablation separates parameter estimation from predictive gain.

Beyond this modest accuracy gain, the value of physics lies in access to a stable physiological parameter and informative mechanistic states.

Arms A5 and A6. Two planned arms were dropped for technical reasons:

- **A5 (simulation pretraining)** would have pretrained on UVA/Padova simulator traces, but the legacy meal generator re-sampled meal times inside the integration loop, producing about 0.6 instead of 3 meals per day; A5 was blocked to avoid training on invalid synthetic data.
- **A6 (legacy heuristic kernels)** would have compared Bergman-derived IOB/COB with legacy kernels that had an unnormalized carbohydrate decay and a time-reversed insulin curve. Instead, the value of the compartmental states is quantified by feature attribution (Section III-J), where they carry 27.1% of total attribution.

1) Multi-Seed Replication and Statistical Significance: As a post hoc check of seed sensitivity, A0 and A7 were each retrained from scratch with five seeds $\{42, 101, 202, 303, 404\}$ under the official protocol. We report the mean and SD of MAE and RMSE across seeds, and paired two-sided Wilcoxon signed-rank tests on seed-averaged per-subject errors with Holm–Bonferroni correction across the four horizons, applied

TABLE VII
ABLATION MATRIX: OFFICIAL PROTOCOL, ALL 12 SUBJECTS, ONE TRAINING RUN PER ARM (SEED 42). MAE AT ALL HORIZONS; CLARKE A AND HYPOGLYCEMIA METRICS AT 30 MIN; HYPO BIAS IS THE MEAN ERROR FOR GLUCOSE BELOW 70 MG/DL. A3 AND A4 DIFFER ONLY IN PER-PATIENT PARAMETERIZATION; PHYSICS WEIGHTING AND CURRICULUM ARE HELD FIXED.

Arm	Configuration	MAE30	MAE60	MAE90	MAE120	Clarke A	Hypo sens.	Hypo bias	Gives S_I
A0	Transformer, no physics	13.51	22.82	29.74	34.93	89.4%	0.621	+7.29	no
A1	Physics penalty, fixed $\lambda = 0.1$	13.68	23.47	30.93	36.22	88.9%	0.488	+11.29	no
A2	Adaptive penalty, per patient	13.58	23.35	30.59	35.74	89.0%	0.486	+11.03	yes
A3	Hybrid prior, curriculum	13.65	23.12	30.02	34.83	89.0%	0.471	+10.95	yes
A4	Hybrid prior, curriculum, population parameters	13.58	23.35	30.57	35.59	89.1%	0.504	+10.18	no
A7	Hybrid, full physics from epoch 1	**13.25**	**22.29**	**28.77**	**33.38**	**89.8%**	0.591	+8.61	yes

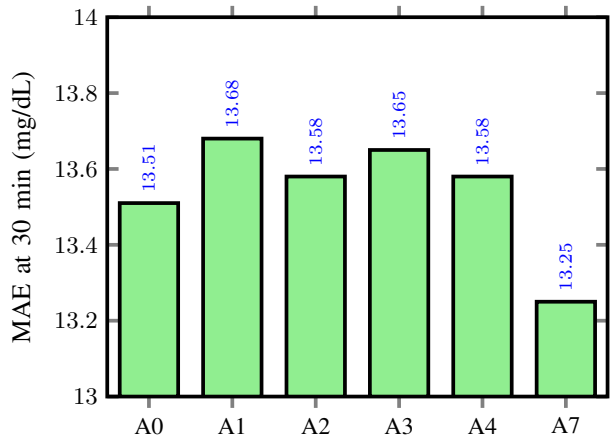


Fig. 5. *Point-accuracy ablation at 30 minutes comparing six physics configurations (A0-A4,A7), from no physics (A0) to full physics from epoch 1 (A7). A7 has the lowest MAE (13.25 mg/dL) versus 13.51 mg/dL for A0 in the primary run (seed 42). A post hoc 5-seed replication (30-min MAE 13.37 vs. 13.54 mg/dL) found A7 modestly but significantly better in MAE at all horizons under paired Wilcoxon testing (Table VIII).*

separately to MAE and RMSE (Table VIII).

A7 had significantly lower MAE than A0 at every horizon ($p_{\text{Holm}} \leq 0.010$), by 0.17 mg/dL at 30 min (1.3% of A0's MAE) rising to 0.99 mg/dL at 120 min (2.8%); subject-bootstrap 95% CIs (5,000 resamples) excluded zero at every horizon ($[-0.30, -0.06]$ at 30 min, $[-0.67, -0.18]$ at 60 min, $[-1.11, -0.37]$ at 90 min, $[-1.49, -0.53]$ at 120 min). The gap grew with horizon in 10 of 12 subjects, although the pre-specified test of this trend was not significant after Holm correction ($p = 0.17$). Physics did not reduce variability across seeds: the seed SD of A7 was similar to A0's at 30 min (0.11 vs. 0.10 mg/dL) and larger at 120 min (0.64 vs. 0.10 mg/dL).

Where the gain arises. In secondary analyses (Section II-H; full tables in the Supplementary Material, Tables S6 and S7), the in-cohort gain concentrated in hyperglycemic targets ($>$ 180 mg/dL: 0.85 mg/dL at 60 min and 1.58 mg/dL at 90 min, both Holm-significant in a family of 55 tests) and in periods of stable glucose ($|\text{ROC}| < 1$ mg/dL/min: 0.78 and 1.12 mg/dL at 90 and 120 min), with no difference in the 70–180 mg/dL range or during rapid change. In that family the overall gaps themselves did not survive correction (Holm $p = 0.07$–0.35), so these strata indicate where the effect lies rather than add independent evidence for it. Contrary to expectation, the gain was not larger in the 3 h after meals (it was, if anything, smaller). With 25% of training subjects, A7's advantage rose to 0.49–2.35 mg/dL across horizons (50%: 0.15–1.41 mg/dL), but with three seeds no low-data comparison survived Holm correction. Forecast trajectories of A0 more often exceeded 4 mg/dL/min (36.8% of windows vs. $< 0.01\%$ for A7), but almost all of these excursions lay in the first and last 10 min of the 2-h spline (0.1% in between), so this reflects smoother horizon boundaries rather than more plausible interior dynamics.

G. Clinical Accuracy and Hypoglycemia Detection

Clinical agreement remains high across horizons: Clarke A+B decreases from 99.0% at 30 minutes to 94.3% at 120 minutes (Fig. 6). The Parkes error grid, pooled across test windows, shows the same pattern: zone A+B falls from 99.8% at 30 minutes to 95.9% at 120 minutes, and zone E is 0% at every horizon.

Because Clarke and Parkes grids were designed for point-in-time sensor measurements rather than forecasts, we also applied Prediction Error Grid Analysis (PRED-EGA) [70], which classifies predictions as Accurate, Benign, or Erroneous separately in hypoglycemia ($\leq$ 70 mg/dL, following the PRED-EGA definition), euglycemia (70–180 mg/dL), and hyperglycemia ($>$ 180 mg/dL) (Table IX). Clinically acceptable predictions (Accurate + Benign) were at least 99.4% at every horizon, and Erroneous predictions were at most 0.6%, reaching that value at 120 min. In hyperglycemia, 96.2% of predictions were Accurate at 30 min and 86.1% at 60 min; in hypoglycemia, Erroneous predictions were 0.00% at 30 min and 0.14% at 60 min.

However, as observed with Clarke zone A, standard point-forecast metrics reward predictions close to the center and can underestimate acute excursions. Thus, for the final model at 30 minutes, 89.8% of predictions in Clarke zone A (mean of per-subject values; Fig. 6) and 99.96% PRED-EGA clinically acceptable can still coexist with the median point forecast missing 45% of hypoglycemic readings (pointwise sensitivity 0.551; Table X). Within the hypoglycemic range itself, only 65.9% of PRED-EGA classifications were Accurate at 30 min, falling to 10.9% at 120 min (Table IX).

H. Hypoglycemia Detection and Predictive Distribution Calibration

Point forecasts trained with squared or Huber losses regress toward the mean and underestimate hypoglycemic excursions. We therefore evaluated hypoglycemia detection in five ways: consensus event-level detection, a comparison with simple

TABLE VIII

POST HOC MULTI-SEED REPLICATION (5 SEEDS: 42, 101, 202, 303, 404) COMPARING A0 (NO PHYSICS) AND A7 (HYBRID PHYSICS-GUIDED), WITH PAIRED WILCOXON SIGNED-RANK TESTS ON SEED-AVERAGED PER-SUBJECT MAE AND HOLM CORRECTION OVER THE FOUR HORIZONS. MEANS ± SD OVER THE FIVE SEEDS. FOR RMSE, THE DIFFERENCE WAS SIGNIFICANT ONLY AT 90 AND 120 MIN ($p_{\text{HOLM}} = 0.027, 0.028$; 30 AND 60 MIN: 0.154).

	A0 (Transformer, no physics)		A7 (Hybrid physics-guided)		Paired Wilcoxon Test, MAE (Seed-Averaged)			
Horizon	MAE (mean ± SD)	RMSE (mean ± SD)	MAE (mean ± SD)	RMSE (mean ± SD)	MAE Diff	p_{raw}	p_{Holm}	Significant?
30 min	13.54 ± 0.10	19.32 ± 0.13	$\mathbf{13.37 \pm 0.11}$	$\mathbf{19.17 \pm 0.16}$	-0.17	0.0093	0.0098	**Yes** ($p < 0.01$)
60 min	23.01 ± 0.16	31.64 ± 0.15	$\mathbf{22.60 \pm 0.32}$	$\mathbf{31.23 \pm 0.44}$	-0.41	0.0049	0.0098	**Yes** ($p < 0.01$)
90 min	30.01 ± 0.15	40.43 ± 0.20	$\mathbf{29.28 \pm 0.53}$	$\mathbf{39.61 \pm 0.70}$	-0.72	0.0024	0.0073	**Yes** ($p < 0.01$)
120 min	35.07 ± 0.10	46.54 ± 0.26	$\mathbf{34.08 \pm 0.64}$	$\mathbf{45.47 \pm 0.86}$	-0.99	0.0015	0.0059	**Yes** ($p < 0.01$)

TABLE IX

PRED-EGA (SIVANANTHAN ET AL. [70]) CLINICAL PREDICTION ERROR GRID RESULTS ACROSS 12 SUBJECTS (MEAN ± SD ACROSS SUBJECTS)

Metric	30 min	60 min	90 min	120 min
Overall Accurate (%)	90.8 ± 3.3	77.8 ± 7.0	68.6 ± 8.1	61.9 ± 8.3
Overall Benign (%)	9.2 ± 3.3	21.9 ± 6.8	30.9 ± 7.8	37.5 ± 8.1
Overall Erroneous (%)	0.04 ± 0.07	0.28 ± 0.34	0.50 ± 0.65	0.60 ± 0.63
Clinically Acceptable (%)	$\mathbf{99.96 \pm 0.07}$	$\mathbf{99.72 \pm 0.34}$	$\mathbf{99.50 \pm 0.65}$	$\mathbf{99.40 \pm 0.63}$
Hypoglycemia Accurate (%)	65.9 ± 28.4	33.8 ± 25.5	18.3 ± 21.3	10.9 ± 16.2
Hypoglycemia Erroneous (%)	0.00 ± 0.00	0.14 ± 0.47	0.14 ± 0.32	0.88 ± 1.61
Hyperglycemia Accurate (%)	96.2 ± 2.0	86.1 ± 6.0	75.6 ± 8.6	66.1 ± 10.4
Hyperglycemia Erroneous (%)	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00

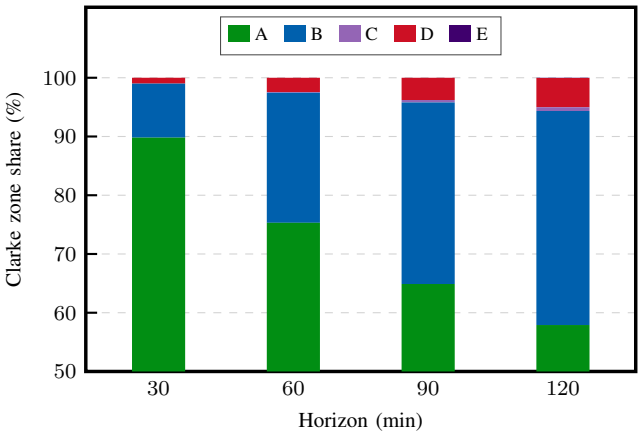


Fig. 6. *Clarke error-grid composition across 30/60/90/120-minute horizons. Zone A decreases from ∼90% to ∼58%, mostly to zone B; zone D rises from about 1% to about 5%, and zone E stays below 0.05%.*

threshold alarms, a five-seed comparison of A0 and A7 with quantile heads, operating curves, and multi-horizon calibration.

1) Consensus Event-Based Hypoglycemia Evaluation: Following international consensus [57], a hypoglycemic episode was defined as at least 15 continuous minutes (three consecutive 5-minute readings) below 70 mg/dL; the test set contains 56 such episodes over 92.0 patient-days (Table X). With 56 events and about 552 effectively independent test windows (494 using the autocorrelation time), these estimates carry substantial uncertainty. The median forecast detected 11 of 56 events (sensitivity 19.6%, specificity 99.4%, 0.62 false alarms/day, median lead time 15 min). An alarm on the lower quantile ($\hat{G}_{0.10} < 70$ mg/dL) detected 38 of 56 before onset (67.9%; specificity 95.4%, 2.21 false alarms/day, median lead time 15 min). Monitoring $\min(\hat{G}_{0.10}(30), \hat{G}_{0.10}(60)) <$ 70 mg/dL detected 46 of 56 (82.1%; specificity 92.1%, 3.46 false alarms/day) and extended the median lead time to 25.0 min. To test whether the quantile alarm adds value over a trivial rule, persistence and linear-extrapolation alarms were given thresholds tuned on the validation split to match the model alarm's validation false-alarm rate (2.23/day), then evaluated once on test. The persistence alarm ($G(t) < 90.5$ mg/dL) detected 38 of 56 events (67.9%, the same as the $q = 0.10$ alarm) at 2.21 false alarms/day with a longer median lead time (17.5 min), but lower window-level specificity (92.0% vs. 95.4%); linear extrapolation (threshold 63.0 mg/dL) detected 25 of 56 (44.6%; specificity 97.4%, 1.96 false alarms/day). At this operating point the quantile alarm therefore does not detect more events than a tuned persistence rule; its advantages are higher specificity, an adjustable operating curve, and the multi-horizon alarm, for which no persistence analogue was evaluated. At about two false alarms per day, alarm fatigue is a concern [78], so the operating point would need to be set with patients.

2) Quantile Alarm for Baseline A0 vs. Physics A7: To test whether physics changes hypoglycemia detection when both models have quantile heads, A0 and A7 were each trained with the same non-crossing quantile head and identical configuration under five seeds, and the $q = 0.10$ alarm was evaluated on the same 56 test episodes (Table XI). Event-level sensitivity was $73.2 \pm 6.7\%$ for A7 and $67.5 \pm 6.7\%$ for A0, at the same specificity (95.3% vs. 95.4%) and false-alarm rate (2.24 vs. 2.22 per day); A0's median lead time was longer (17.0 vs. 14.0 min). The difference in sensitivity (+5.7 percentage points) had a 95% CI of $[-5.7, +15.3]$ from a two-level bootstrap over subjects and seeds, so physics did not significantly change hypoglycemia detection. Single seeds varied widely (60.7–80.4%), and a single-seed comparison could point in either direction; an earlier single-seed comparison is superseded by this replication.

Single-run hypoglycemia estimates are also sensitive to training. In the single-seed ablation without a quantile head (Table VII), A7 had lower sensitivity than A0 (0.591 vs. 0.621) and a larger hypoglycemic bias (+8.61 vs. +7.29 mg/dL). Tables VII and X therefore report different runs and architectures and should not be compared directly.

3) Alarm Operating Curves: The 70 mg/dL alarm threshold was fixed in advance from the consensus definition [57] rather than tuned; on test it gave pointwise sensitivity 0.918, specificity 0.951, and precision 0.347 (1,257 false positives). Only $q \in \{0.1, 0.5, 0.9\}$ are trained (Eq. (16)); the levels $q = 0.05$ and 0.20 were obtained by treating the lower half of the predictive distribution as Gaussian, with $\sigma_L = (\hat{G}_{0.5} - \hat{G}_{0.1})/(-\Phi^{-1}(0.10))$ and $\hat{G}_p = \hat{G}_{0.5} + \sigma_L \, \Phi^{-1}(p)$, an approximation rather than independently trained quantiles. Sensitivity–precision curves for $q \in \{0.05, 0.10, 0.20\}$ (in the code repository) show that the operating point can be adjusted along a continuous trade-off.

4) Calibration and Uncertainty Quantification: Table XII reports calibration, sharpness, and scoring rules. Empirical coverage of the nominal 80% interval ($[q_{0.10}, q_{0.90}]$) was 79.4%, 78.5%, 77.2%, and 76.9% at 30, 60, 90, and 120 min, and the lower quantile covered 9.47% of observations at 30 min and 9.96% at 120 min, close to the nominal 10.0%. The upper quantile, in contrast, was under-covered at every horizon: 88.9%, 88.1%, 87.4%, and 86.9% of observations fell below it at 30, 60, 90, and 120 min, against a nominal 90%. For glucose below 80 mg/dL, the interval covered 76.4% of observations and the lower quantile 22.4%. Between 30 and 120 min, mean interval width grew from 39.9 to 96.7 mg/dL and CRPS from 8.43 to 21.01 mg/dL. Split-conformal recalibration (a per-horizon shift of $q_{0.10}$ and $q_{0.90}$ fitted on validation data) restored near-nominal marginal coverage of the 80% interval (82.3% at 30 min, 79.5% at 120 min) but not coverage below 80 mg/dL (78.1% and 51.9%), so the miscalibration depends on glucose level and would need glucose-conditional recalibration. A per-subject linear quantile regression on the same 35 features had slightly higher CRPS (8.54 to 22.75 mg/dL) and wider intervals (43.8 to 114.0 mg/dL). With only three trained quantiles, CRPS was approximated by trapezoidal integration of twice the pinball loss over $\tau \in \{0, 0.1, 0.5, 0.9, 1\}$ (zero at the ends): $0.5\,\text{pinball}_{0.1} + 0.8\,\text{pinball}_{0.5} + 0.5\,\text{pinball}_{0.9}$.

Point forecasts also increasingly compress glucose excursions at longer horizons. The predicted-to-actual coefficient-of-variation ratio decreases from 0.973 to 0.802, with TIR increasingly overestimated and TBR underestimated (Fig. 7). Consistently, predicted LBGI falls increasingly below the actual value, from 0.65 versus 0.77 at 30 minutes to 0.33 versus 0.74 at 120 minutes, indicating progressively weaker representation of hypoglycemic risk in longer-horizon point forecasts.

I. Patient-Specific Insulin Sensitivity and Validity Checks

Three pre-specified checks determine whether S_I may be interpreted physiologically (Table XIII). All pass under the official protocol: between-subject variation is non-degenerate (CV $= 35.8\%$), test–retest reliability is high (ICC$(1) = 0.890$), and S_I is strongly inversely related to total daily insulin dose ($\rho = -0.839$, 95% CI $[-0.99, -0.44]$). Under LOSO, stability and variability remain, but the dose correlation fails. Hence S_I is a stable, subject-specific estimate whose dose relation holds only in the personalized setting. Because the parameter head sees the insulin inputs and total daily dose is computed from the same records, this relation is concurrent consistency, not external validity; cross-subject validity is not established.

For the stability check, each subject's window-level S_I estimates were split into four disjoint, time-ordered blocks, each reduced to its median, and ICC(1) was computed across subjects on these four block medians. Each subject's S_I is the median of its window-level estimates; the between-subject CV and the dose correlation are computed across these 12 subject medians. Total daily dose was the total recorded insulin delivery divided by the number of recorded days (U/day).

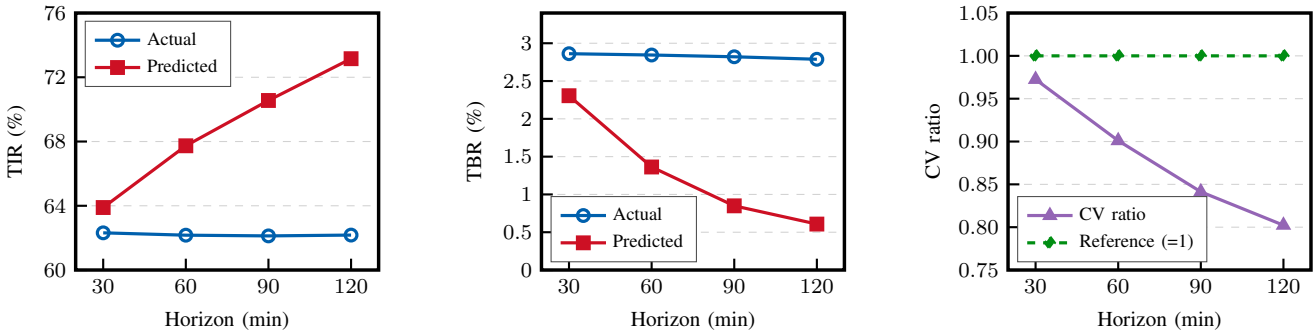
1) Addressing Potential Circularity and Confounding: We tested whether the stability and dose correlation of S_I could be artifacts of the temporal-consistency loss or of ambient glucose, and whether S_I agrees with classical minimal-model estimation (Table XIV):

1) **No temporal-consistency loss ($\lambda_{tc} = 0$):** Retrained without the penalty, S_I still passed all checks: ICC(1) = 0.645, CV = 22.2%, all estimates within the Ward IDDM bounds, and $\rho = -0.804$ with total daily dose ($p = 0.0016$, 95% CI $[-0.955, -0.451]$). Stability is therefore partly, but not solely, attributable to the penalty: removing it lowered ICC(1) from 0.890 to 0.645 and CV from 35.8% to 22.2%.
2) **Partial correlation controlling for mean glucose:** Controlling for test-interval mean glucose, the S_I–TDD partial correlation was $r = -0.838$ (rank, $p = 0.0013$) and $r = -0.885$ (Pearson, $p = 0.0003$); controlling for whole-study mean glucose gave $r = -0.843$ ($p = 0.0011$) and $r = -0.875$ ($p = 0.0004$) (partial correlations computed from OLS residuals, p from a t test with $n - 3$ df). Mean glucose does not explain the relationship.
3) **Independent minimal-model fit:** S_I from the parameter head agreed with per-subject Bergman estimates obtained by nonlinear least squares/MAP (Spearman $\rho = 0.790$, $p = 0.0022$; Pearson $r = 0.831$, $p = 0.0008$), without iterative fitting at inference. This reference is weak (30-min MAE 25.5 mg/dL even on training data), so agreement indicates only loose convergent validity.
4) **Insulin-to-carbohydrate ratio:** For the six 2018 subjects with clinical ICR (grams of carbohydrate per unit of insulin, so a positive correlation is expected), the correlation with S_I was Spearman $\rho = +0.371$ ($p = 0.469$) and Pearson $r = +0.079$ ($p = 0.881$); with $n = 6$, this check is underpowered.
5) **Learned trust gate g:** The gate on the Bergman prior, initialized at $g_0 \approx 0.10$, converged to 0.152 in the point forecaster (A7) and 0.128 in the quantile forecaster.

Body weight is not identifiable because all OhioT1DM files use the same de-identification placeholder; a fixed nominal

TABLE X
CONSENSUS EVENT-LEVEL AND POINTWISE HYPOGLYCEMIA DETECTION AT 30 MINUTES FOR THE FINAL MODEL AND VALIDATION-TUNED THRESHOLD ALARMS (92.0 PATIENT-DAYS, 56 EPISODES). THE FIVE-SEED A0 VS. A7 COMPARISON IS IN TABLE XI.

Model	Alarm Head	Point Sens.	Point Spec.	Point Prec.	Event Sens.	Specificity	False Alarms/Day	Median Lead Time
A7 (Physics hybrid)	Median (point)	0.551	0.993	0.699	19.6%	99.4%	0.62	15.0 min
A7 (Physics hybrid)	Quantile $q = 0.10$	0.918	0.951	0.347	67.9%	95.4%	2.21	15.0 min
A7 (Physics hybrid)	Multi-horizon $q = 0.10$	–	–	–	**82.1%**	92.1%	3.46	**25.0 min**
Persistence	$G(t) < 90.5$ (val.-tuned)	–	–	–	67.9%	92.0%	2.21	17.5 min
Linear extrapolation	$\hat{G} < 63.0$ (val.-tuned)	–	–	–	44.6%	97.4%	1.96	15.0 min



(a) **(b)** **(c)**

Fig. 7. *Excursion compression with horizon (30/60/90/120 minutes). a: actual time-in-range (TIR) stays flat near 62% while predicted TIR climbs toward 74%, an increasing overestimate. b: actual time-below-range (TBR) stays flat near 2.8% while predicted TBR falls toward 0.6%, an increasing underestimate of hypoglycemic exposure. c: the predicted-to-actual coefficient-of-variation ratio declines from about 0.97 to 0.80. All three panels show the same mechanism: as the horizon lengthens, the point forecast increasingly regresses toward the mean and loses the variance of the true glucose trace.*

TABLE XI
FIVE-SEED HYPOGLYCEMIA ALARM ($\hat{G}_{0.10} < 70$ MG/DL, 30 MIN) FOR A0 AND A7 WITH IDENTICAL QUANTILE HEADS AND TRAINING CONFIGURATION; MEAN ± SD OVER SEEDS 42, 101, 202, 303, 404 (56 EPISODES). DIFFERENCE CI FROM A TWO-LEVEL BOOTSTRAP (SUBJECTS, THEN SEEDS; 5,000 RESAMPLES).

Model	Event Sens.	Specificity	FA/day	Lead (min)
A0	$67.5 \pm 6.7\%$	$95.4 \pm 0.4\%$	2.22 ± 0.19	17.0 ± 2.1
A7	$73.2 \pm 6.7\%$	$95.3 \pm 0.4\%$	2.24 ± 0.16	14.0 ± 2.2
A7 − A0	+5.7 pp, 95% CI $[-5.7, +15.3]$ (not significant)			

body weight of 70 kg is therefore used for every subject wherever weight is required (the per-kilogram volumes V_G and V_I and the glucose appearance rate R_a). The dose-per-kilogram correlation is therefore not independent of total daily dose. Moreover, p_2 and p_3 are range-constrained, so agreement with a plausible interval is partly imposed rather than learned.

J. Feature Attribution

IG and permutation importance agree with Spearman $\rho = 0.740$ ($p = 3.83 \times 10^{-7}$, $n = 35$) and share three of their top five features. Both rank five-minute glucose rate of change first by a wide margin. Permuting it raises MAE by 3.11 mg/dL, over seven times the next feature. The group-level IG result in Fig. 8 assigns 42.7% to glucose-derived features and 27.1% to mechanistic states, suggesting that physiology-derived inputs carry information, although attributions among correlated features should be read with caution.

Attribution concentrates in the most recent input positions, supporting attention pooling. The IG completeness discrepancy has median relative magnitude 3.2% and p_{95} 70%; it does not shrink with more integration steps and remains unresolved. Conclusions are therefore restricted to findings corroborated by permutation importance.

K. Comparison With Published Work

At 30 minutes, the current work is mid-field relative to the 2020 challenge literature [66], [71-74], [75] (Table I): on the matched 2020 cohort (subjects 540/544/552/567/584/596, the six subjects on which the challenge entries were scored), RMSE is 18.74 and MAE is 13.24 mg/dL (20.93 and 14.98 mg/dL under LOSO). On the 2018 cohort, RMSE@30 is 18.93 mg/dL, lower than four of the five 2018 entries but above [37] (17.80 mg/dL). A further 2020 entry reports RMSE@30 of 19.60 mg/dL [76]. At 60 minutes, RMSE (31.16) is essentially tied with the best entry (31.10), and MAE (22.55) is the second lowest; differences in window eligibility, missing-data handling, external pretraining, and best-configuration reporting prevent a state-of-the-art claim. Across all 12 subjects, the only published comparison is GARNN [77] (18.97/13.34 mg/dL at 30 minutes), against which the all-subject result (18.84/13.08) should be read as a tie. The defensible advances are an evaluation of hypoglycemia alarms against a simple baseline, explicit personalized-versus-LOSO decomposition, pre-specified checks of a physiological parameter, and a quantified account of the benefits and costs of Bergman constraints. The all-12-subject figures in the

TABLE XII
MULTI-HORIZON CALIBRATION, CONDITIONAL COVERAGE, INTERVAL WIDTH, AND SCORING RULES ACROSS ALL 12 SUBJECTS (26,498 TEST WINDOWS)

Horizon	Nominal 80% Cov.	Lower $q_{0.10}$ Cov.	Upper $q_{0.90}$ Cov.	Cov. ($G < 80$)	Width (mg/dL)	Pinball ($q_{0.10}$)	Pinball ($q_{0.50}$)	CRPS (mg/dL)
30 min	79.4%	9.47%	88.9%	76.4%	39.9	2.98	6.53	8.43
60 min	78.5%	9.63%	88.1%	64.1%	65.9	4.77	10.96	13.99
90 min	77.2%	10.22%	87.4%	55.7%	83.2	6.02	14.17	18.02
120 min	76.9%	9.96%	86.9%	53.3%	96.7	6.94	16.57	21.01

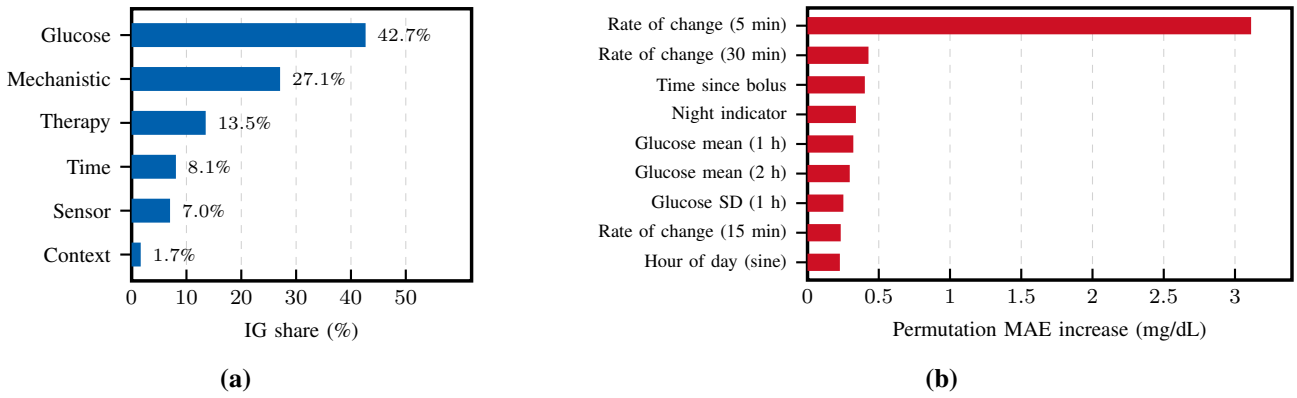


Fig. 8. *Feature attribution at 30 min using integrated gradients (a) and permutation importance (b). Glucose-derived features dominate IG attribution (42.7%), while the 5-min glucose rate of change is the strongest permuted feature (∼3.1 mg/dL MAE increase).*

TABLE XIII
PRE-SPECIFIED CHECKS OF INSULIN SENSITIVITY

Check	Threshold	Official	LOSO
Between-subject CV	>10%	35.8%	34.3%
Test–retest ICC(1)	>0.5	0.890	0.816
ρ vs. total daily dose	<0, CI excludes 0	−0.839	−0.357
95% CI for ρ	–	[−0.99, −0.44]	[−0.82, 0.28]

Results (e.g., 18.84/13.08 at 30 minutes) differ from the per-cohort figures in Table I because they answer a different comparability question.

IV. DISCUSSION

The results separate three concepts that are often conflated in digital-twin and glucose-prediction literature [2]. First, a mechanistic constraint can be useful beyond point accuracy. Here it improves in-cohort accuracy only modestly (0.17–0.99 mg/dL MAE across horizons over five seeds, 1.3–2.8%), too little to change a clinical decision on its own. The gain is larger where a physiological prior has most leverage: at longer horizons, in hyperglycemia, and, most consistently, under distribution shift, where models trained on one OhioT1DM cohort and tested on the other were 1.7–7.4 mg/dL (about 9–17%) more accurate with physics in every test subject. This pattern is consistent with physics acting as a regularizer when training data do not match the test population, which is the situation a deployed forecaster faces; confirming it requires a genuinely external cohort. It also supplies interpretable state variables and a stable S_I estimate, and mechanistic features account for more than one quarter of total IG attribution. Second, a favorable error grid does not demonstrate event safety. The median forecast has excellent Clarke A+B coverage but inadequate hypoglycemia sensitivity; a lower-quantile alarm detects more events than the median forecast, but no more than a persistence alarm tuned to the same false-alarm rate, and its calibration below 80 mg/dL is poor. Physics did not change alarm performance. Forecast-based alarms in this cohort should therefore be justified by specificity, adjustable operating points, and multi-horizon warning, not by event sensitivity alone. Third, personalized temporal validation and cross-subject generalization answer different questions. The growing official–LOSO gap shows that access to a subject's history matters increasingly at longer horizons. Prospective digital-twin trials demonstrate the longer-term potential of individualized replay and adaptation, but do not remove the need to validate the forecasting layer independently [17].

The ablation also exposes a structural limitation. Bergman's equilibrium at G_b pulls trajectories toward the center of the glucose distribution. In the best configuration this adds approximately 1.3 mg/dL to an already-present regression-to-the-mean bias during hypoglycemia. This behavior appears whether physics enters as a penalty or a prior. A future model should modify the physiological structure rather than hide the bias with an asymmetric point loss.

V. LIMITATIONS AND FUTURE WORK

The cohort contains only 12 subjects and approximately 2,800 effectively independent observations, of which only about 550 fall in the test set; the hypoglycemia analysis rests on 56 events. A tuned persistence alarm matched the quantile alarm's event sensitivity. The ablation screen used a single seed (42); only A0 and A7 were replicated over five seeds, post hoc (Section III-F1). The secondary analyses were specified after the main results were known; the low-data comparison used three seeds and the cross-cohort comparison six subjects per direction, so neither reached significance after correction, and the cross-cohort test is transfer within one dataset rather than

TABLE XIV
NON-CIRCULARITY, CONFOUNDING, AND PHYSIOLOGICAL VALIDATION OF LEARNED INSULIN SENSITIVITY S_I

Analysis / Validation Check	Value	Interpretation / Threshold
Temporal Consistency Ablation ($\lambda_{tc} = 0.0$):		
Between-subject CV	22.2%	Non-degenerate ($> 10\%$)
Test–retest reliability ICC(1)	0.645	Stable without penalty (> 0.5)
Spearman ρ vs. TDD	-0.804	Significant ($p = 0.0016$, 95% CI $[-0.96, -0.45]$)
Ward IDDM range adherence	100%	Within bounds (partly imposed by design)
Confounding by Ambient Glucose:		
Unadjusted $\rho(S_I, \text{TDD})$	-0.839	Baseline ($p = 0.0006$)
Partial rank $r(S_I, \text{TDD} \mid \bar{G}_{\text{test}})$	-0.838	After adjustment ($p = 0.0013$)
Partial Pearson $r(S_I, \text{TDD} \mid \bar{G}_{\text{test}})$	-0.885	After adjustment ($p = 0.0003$)
Partial rank $r(S_I, \text{TDD} \mid \bar{G}_{\text{all}})$	-0.843	After adjustment ($p = 0.0011$)
Agreement with Classic Minimal Model:		
Agreement with NLS/MAP Bergman (Spearman ρ)	$+0.790$	Weak reference ($p = 0.0022$)
Agreement with NLS/MAP Bergman (Pearson r)	$+0.831$	Weak reference ($p = 0.0008$)
Agreement with clinical ICR ($n = 6$, exploratory)	$+0.371$	Underpowered ($p = 0.469$)
Mechanistic Prior Trust Gate:		
Initialized gate $g_0 = \sigma(-2.2)$	9.98%	Initialization
Converged gate g (Point Forecaster, A7)	15.2%	Converged value
Converged gate g (Quantile Forecaster, A7)	12.8%	Converged value

external validation. A7 was more variable across seeds than A0 at longer horizons (not significant). The evidence for S_I is internal: it is stable and dose-consistent within this cohort, with only a weak external reference (Section III-I); OhioT1DM has no clamp measurement, the dose relation fails under LOSO, and the usefulness of S_I, e.g., for insulin dosing, has not been tested. Subject 552 has only 59.7% test CGM coverage, and subject 567 has no carbohydrate records in the test interval. Body weight is unavailable, so a fixed 70 kg is assumed for every subject. The alarm threshold was fixed at the consensus 70 mg/dL rather than chosen from an elicited clinical cost ratio (Section III-H3). No counterfactual intervention is validated, so the system is a physics-guided forecaster rather than a digital twin or treatment simulator.

The parameter head estimates a ten-component physiological vector θ_p for each subject, rather than only S_I. Table XV lists its components: p_1, p_2, p_3, n, V_G, V_I, $t_{\max,I}$, k_{gri}, k_{abs}, and f.

Pre-specified validity checks (Table XIII and Table XIV) were carried out for S_I alone, for three concrete reasons rather than by omission. First, S_I is the one component with a direct clinical measurement analogue: a euglycemic clamp measures exactly this quantity, giving a real external target to validate against, whereas the remaining parameters have no equivalent bedside measurement available in OhioT1DM. Second, S_I has literature-sourced plausible bounds specific to insulin-dependent diabetes [44], giving its sigmoid-mapped range a population anchor that is not equally well established for every other compartment parameter. Third, the ablation (Table VII, A3 vs. A4) already shows that per-patient parameter estimation is accuracy-neutral overall (0.07–0.76 mg/dL), so the contribution claimed here is the stability and dose consistency of one physiological quantity, not evidence that the other nine are unimportant. One concrete obstacle already blocks part of this extension: body weight is unavailable because every OhioT1DM file carries the same de-identification placeholder, which is why a body-weight-normalized analysis was not attempted for any parameter, not only S_I.

Priority extensions are external validation on larger cohorts (e.g., Shanghai, DiaTrend), formal identifiability analysis of the remaining patient parameters, pre-specified validation of the remaining Bergman and absorption parameters (p_1, n, V_G, V_I) against literature-sourced population anchors comparable to those used for S_I, time-varying S_I, a joint multi-horizon predictive distribution, decision-theoretic alarm selection with patient-specific cost matrices, longer context, and prospective validation of counterfactual meal and insulin interventions.

VI. CONCLUSION

In this 12-subject cohort, an integrity-checked, hybrid physics-guided forecaster outperformed a validated persistence baseline by 19–21% RMSE across 30–120-minute horizons. Physics improves in-cohort point accuracy only modestly, and significantly only in a post hoc five-seed replication, but it degraded less when models were transferred between cohorts. The strongest evidence is that physiology can provide stable patient-specific structure alongside this small accuracy gain. Lower predictive quantiles recovered hypoglycemic events missed by the median forecast, but no more than a tuned persistence alarm, and physics did not change alarm performance. Horizon integrity, subject-level statistics, matched generalization protocols, and explicit safety trade-offs are prerequisites for credible glucose forecasting.

TABLE XV
PATIENT-PARAMETER VECTOR θ_p AND ADMISSIBLE INTERVALS

Symbol	Meaning	Range	Unit	Population anchor
p_1	Insulin-independent glucose disposal	0–0.030	min^{-1}	0.013
p_2	Decay of remote insulin action	0.005–0.10	min^{-1}	0.025
p_3	Build rate of action from insulin excess	10^{-6}–3×10^{-5}	mL $\mu\text{U}^{-1}\text{min}^{-2}$	6.25×10^{-6}
n	Plasma insulin clearance	0.08–0.25	min^{-1}	0.138
V_G	Glucose distribution volume	1.4–2.4	dL/kg	1.88
V_I	Insulin distribution volume	0.08–0.18	L/kg	0.12
$t_{\max,I}$	Time to peak SC absorption	30–90	min	55
k_{gri}	Gastric emptying rate	0.008–0.10	min^{-1}	0.035
k_{abs}	Intestinal absorption rate	0.005–0.10	min^{-1}	0.0167
f	Carbohydrate bioavailability	0.70–1.00	dimensionless	0.90

APPENDIX A
NOTATION, VARIABLES, AND TERMINOLOGY

This appendix summarizes the notation and feature groups used in the manuscript; full notation tables, including abbreviations, are in the Supplementary Material. Rate constants are expressed per minute unless stated otherwise. A subscript b denotes a basal or fasting quantity, a hat denotes a prediction, bold lowercase symbols denote vectors, and bold uppercase symbols denote matrices.

A. Physiological States and Inputs

Glucose G is propagated separately from a six-state linear compartment model. Insulin on board and carbohydrate on board are derived identities, not additional fitted states (cf. Eqs. (6)–(7), (9)–(10)):

$$\text{IOB} = S_1 + S_2, \qquad \text{COB} = Q_{\text{sto}} + Q_{\text{gut}}. \tag{19}$$

B. Estimated Physiological Parameters

The parameter head emits unconstrained values z_i, which are mapped to admissible, physiologically sourced intervals [44] by

$$\theta_i = \ell_i + (u_i - \ell_i)\sigma(z_i). \tag{20}$$

The bounds prevent nonphysical values but also mean that plausible range alone cannot establish identifiability.

The reported derived quantity is

$$S_I = p_3/p_2, \tag{21}$$

the steady-state gain from plasma insulin above basal to remote insulin action, already introduced as Eq. (5) [1], [43]. Higher S_I means a given insulin excess produces stronger glucose disposal. It is interpreted as a subject-specific estimate only when its cross-subject variation, test–retest ICC, and dose correlation meet the pre-specified criteria.

The implementation-level objective, following Eq. (17), can be written more explicitly as

$$\mathcal{L} = \mathcal{L}_{\text{Huber}} + \lambda_q \mathcal{L}_{\text{pinball}} + \lambda_{\text{phys}} \mathcal{L}_{\text{phys}} + \lambda_{\text{prior}} \mathcal{L}_{\text{prior}} + \lambda_{\text{tc}} \mathcal{L}_{\text{temporal}}. \tag{22}$$

C. Feature Contract

The 35-input feature contract contains six groups: nine glucose features (current glucose, three rates of change, and rolling summaries); nine mechanistic features (IOB, COB, the six compartment states, and glucose appearance); four therapy/timing features; three cyclic-time features; six sensor values or availability masks; and four behavioral/context indicators. Glucose features account for 42.7% of IG magnitude, mechanistic features 27.1%, therapy 13.5%, time 8.1%, sensors 7.0%, and context 1.7%. Every sensor channel has an availability indicator so missingness is not represented as a physiologically extreme standardized zero. Input interpolation is explicitly flagged, whereas forecast targets are always real CGM observations.

Table XV lists the estimated patient-parameter vector. Notation tables for physiological states and inputs, architecture, discretization and uncertainty, training losses, evaluation metrics, and abbreviations are provided in the Supplementary Material.

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